

What a Small Local LLM Can and Cannot Do for Legal Case Research

Evidence from Intellectual Property and Immigration law: a capability-level evaluation of Phi-4-mini-instruct across three independent pipelines

Swetaleena Guha · Nandini Chanda · Nishanth Madthila

Team 4 · 95-820 Applications of NL(X) and LLM · Heinz College, Carnegie Mellon University · Fall 2026 · Instructor: Prof. Sara Kingsley

Research prototype · not legal advice

Federal Court of Australia, 2006–2009

1 Problem & Research Questions

LLMs read long judgments fast, but a fluent answer can cite a case that does not exist. Courts have sanctioned lawyers for filing fabricated citations (*Mata v. Avianca*, 2023).

31% of associates' working hours go to legal research, in a survey of 190 U.S. law firm associates (Lastres, 2013)

The question: can errors be caught before they reach a client or a court?

- RQ1 Capability.** How reliably can a small local model extract area of law, cited cases, legislation, relief and citation treatment?
- RQ2 API features.** Do structured output and tool calling beat plain generation?
- RQ3 Controls.** What do validation and rule-based guards catch, and miss?
- RQ4 Adoption.** What is ready for use, and under what oversight?

2 Data, System & Design

3,890 Federal Court of Australia judgments, UCI Legal Case Reports dataset (Galgani et al., 2012)

IP-A · Guha 214 whole IP case reports	IP-B · Chanda 218 records from 50 IP judgments	IMM · Madthila 220 immigration opinions, 137 with citation tables
---	--	---

Target fields

- ip_area
- legislation_cited
- relief_sought
- cited_cases
- citation_treatment (11)
- visa_category

Model: Microsoft Phi-4-mini-instruct (3.8B), run locally through LLMBBox. No hosted API.

Six capability families, tested in each study

1 Baseline generation	2 Structured output
3 Tool calling	4 Input / output validation
5 Guardrails vs. probes	6 Human adjudication

Three studies, built independently. Corpora, tasks and splits differ, so results are compared side by side and never pooled. Most runs: 50 records.

3 Main Results

R1 Baselines fall short, even of a trivial guess

IMM citation treatment, completed dev records

Always answer "cited"	44%
Phi, temp 1.0	12.5%
Phi, temp 0.2	6.2%

Best IP-B baseline: **52%** exact match on IP area, with **5 fabricated citations** per run. 18 of 50 IMM records ran out of GPU memory.

R2 Structured output fixed form, not content

IP-B, same 50 evaluation records: plain prompt → structured output

Valid JSON 50 / 50	IP exact match 50% → 42%	Made-up citations 1 → 13
------------------------------	------------------------------------	------------------------------------

IP-A got **0 of 50** valid JSON; IMM got **10 of 50** as returned. In both, the model echoed the schema instead of filling it. LLMBBox puts the schema in the prompt; it does not enforce it.

R3 Tool calls failed as the model got more say

Call fully specifiedModel decides

IP-A lookup_case 50 / 50 correct calls; document ID given in the prompt	IP-B check_citations 13 / 50 records skipped the tool; recall fell to 34%	IMM verify_citation 0 calls yet 27 answers said "VERIFIED: true"
---	---	--

R4 Rule-based guards catch what they were written for

Probe type	IP-A	IP-B	IMM
Indirect injection caught	10 / 10	2 / 3	5 / 5
Harmful / toxic caught	0 / 10	0 / 3	4 / 5
Benign requests blocked	0 / 10	1 / 5	0 / 10
Human agreement	80%	76%	96.7%

Guards cost under 3.4 ms per request. Reworded harmful requests slipped past word lists in all three studies. Probe sets differ, so rates are not directly comparable.

4 Patterns Across Both Domains

- No baseline was usable unreviewed**, and decoding changes that helped one study hurt another.
- Valid format ≠ correct content.** Schema-valid output never raised accuracy.
- Discretion hurts reliability.** Tools worked only when the call was spelled out.
- System-run checks work where aimed**, but they match strings, not law, and drop real citations written differently.
- Read the raw outputs.** Schema echoing, OOM records and silently dropped tool calls never showed in aggregates.

HIGHLIGHTED FAILURE CASE

27 answers claimed **"VERIFIED: true"** without a single tool call

Some "verified" items were not even cases: a magistrate's name, a misremembered High Court case. A reviewer trusting the label would get an invented authority marked as checked.

Lesson: only the system can verify. A model's claim of verification carries no weight.

5 Recommendation & Future Work

ADOPT NARROWLY, WITH HUMAN REVIEW

Use for Read-only lookup by known ID Navigating approved material Drafts linked back to the source System-run grounding and checks	Do not use for Unreviewed extraction Citation treatment labels Self-reported verification Guards as the only safety layer
---	--

Cost note. IMM's model saves clinic time only if over 25% of outputs are usable as is; measured: at most 4–8%. Every cost model hinges on review time, which no study measured.

Future work

Pattern-based citation extraction; LLM only for treatment, on the citing paragraph.

Schema-enforced decoding, scoring format and accuracy separately.

Normalised verification of short-form citations, indexes

6 FINAL TAKEAWAY

Let the system, not the model, retrieve and verify, and have a qualified person review every substantive output against the source.

Key references: Galgani, Compton & Hoffmann (2012), UCI Legal Case Reports · Abouelenin et al. (2025), Phi-4-Mini · Kingsley (2026), LLMBBox · Dahl et al. (2024); Magesh et al. (2025), legal hallucination · Lastres (2013) · *Mata v. Avianca* (S.D.N.Y. 2023). Full list in the report.